

Lab 1 – Ideal Gas Law

CHE 260 – Thermodynamics and Heat Transfer

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1 Introduction

1.1 Ideal Gas Law

Gases are a fundamental foundation for understanding chemistry and the quantitative laws of their behavior provide us with a direct connection between macroscopic and particle level properties. A principal example of this is the Ideal Gas Law. The Law was first derived in 1834 by Benoît Paul Émile Clapeyron who combined several empirical laws including Boyle’s Law and Charles’s Law [1]. The most common way to write this law is:

$$PV = NR_uT \quad (1)$$

Where P is the pressure measured in the system, for this experiment the Pressure value is calculated as $P = P_{\text{gauge}} + P_{\text{atm}}$ where P_{gauge} is the pressure measured on the gauge and P_{atm} is the atmospheric pressure of the room the experiment took place in, V is the volume of the system, N is the amount of moles of gas present in the system, R_u is the gas constant, $8.31.4 \text{ J mol}^{-1}\text{K}^{-1}$ and finally T is the temperature of the system expressed in Kelvin. By understanding that N is the quotient of the total mass of the gas in the

system and the molar mass of the gas(s), then the formula can be rearranged to

$$PV = mRT \quad (2)$$

Where the new R constant is the gas constant R_u divided by the molar mass of the gas. For Air this value is $287 \text{ J}/(\text{kg} \cdot \text{K})$ [2]. This is important for this experiment as the mass of the air can be calculated through the measured mass flow rate. In an “ideal gas” as the law describes, particles move stochastically and independently of each other, only interacting through elastic collisions which conserve both momentum and kinetic energy [3].

1.2 Thermodynamic State

Properties and Path Independence

A Thermodynamic property is something which affects the energy of the system that can be measured without knowing the previous state of the system [4]. These properties are used to describe the thermodynamic state of the system, these include pressure, temperature, volume, mass, and density. The properties listed also only depend on the

current state of the system, these are called state properties and the principle is known as path independence.

This concept is especially important for this lab, as the first part compares two different methods of equalizing pressure. In the first method, air is moved rapidly through a solenoid valve, while in the second method, air is transferred gradually through a needle valve. While intermediate sections of the processes may vary, both methods are expected to produce the same final equilibrium conditions.

1.3 Heat Transfer and the First Law of Thermodynamics

This is described through the First Law of Thermodynamics which states that energy is the change in internal energy of a closed system is equal to the heat transferred into the system subtracted by the work done by the system [4]. Because the total volume of the system in the lab is constant the change in internal energy is equivalent to the net heat transfer.

1.4 Air Compressibility

Finally another important characteristic of gases is their compressibility which describes how their volume or density changes in response to variations in pressure. Unlike liquids and solids, gases are highly compressible due to the large distance between molecules [5].

In this experiment the two tanks have a constant volume throughout the process. Because of this the changes in pressure are related to the changes in mass, density, and the temperature of the air in each tank. This relationship can be rearranged to be

$$\rho = \frac{P}{RT} \quad (3)$$

However, the Ideal Gas Law assumes that the volume occupied by individual gas molecules and the intermolecular forces between them are negligible.

2 Results and Discussion

2.1 Part 1: Path Independence and Tank Volume Ratio

Part 1 of the lab discusses the comparison between the movement of air through two tanks with identical volume, where one has a gauge pressure of 40 PSI to another with a gauge pressure of 0 PSI. The main point of comparison is to find the differences in a slow equalization of pressure versus a rapid equalization of pressure, particularly to prove that the volume ratio is close to 1, thus showing that the tanks have identical volume.

2.1.1 Processed Results: Rapid and Slow Expansion

Figures 1–2 show the measured pressure and temperature histories for the rapid and slow expansions.

Taking averages across the marked time ranges on Figures 1–2 for final pressure and temperature values, we obtain the following values:

$$P_{2R} = 24.13 \text{ PSI}, P_{2L} = 24.03 \text{ PSI}$$

$$T_{2R} = 28.33^\circ\text{C}, T_{2L} = 25.25^\circ\text{C}$$

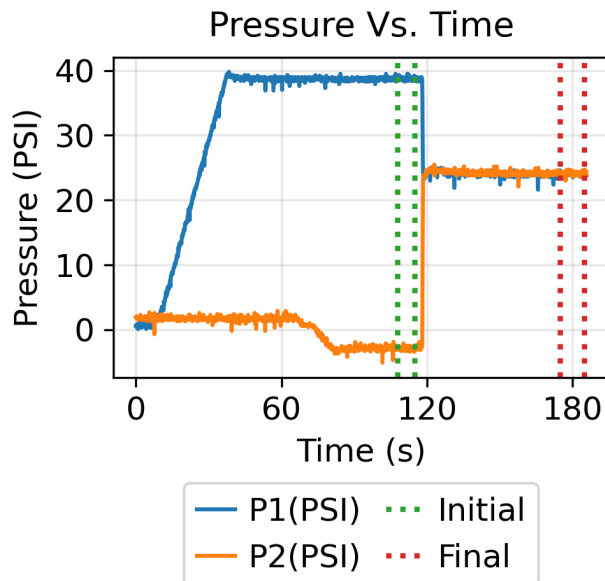


Figure 1: Tank pressures during the Part 1a rapid expansion.

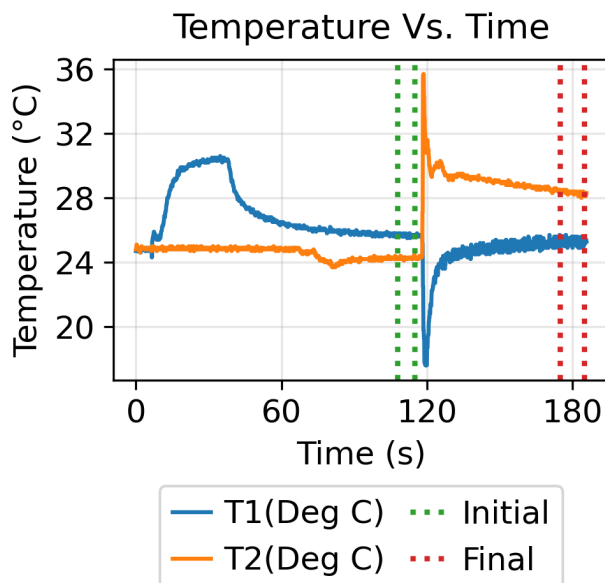


Figure 2: Tank temperatures during the Part 1a rapid expansion.

Using these values with Ideal Gas Law, we can determine the proportionality of pressure, volume and temperature between the two tanks at the final state. It can be assumed that the ratio of the final pressure, volume and temperature of the left tank is equal to the final pressure, volume and temperature of the right tank since both tanks are in equilibrium with each other at the final values. The following calculations show the resultant volume ratio of the rapid equilization of pressure

$$\begin{aligned}\frac{P_{2L}V_L}{T_{2L}} &= \frac{P_{2R}V_R}{T_{2R}} \\ \frac{V_L}{V_R} &= \frac{P_{2R}}{P_{2L}} \cdot \frac{T_{2L}}{T_{2R}} \\ \frac{V_L}{V_R} &= \frac{24.13}{24.03} \cdot \frac{25.25}{28.33} \\ \frac{V_L}{V_R} &= 0.895\end{aligned}$$

Similar to Figures 1–2, the following Figures 3–4 show the pressures and temperatures of both the left and right tank during the slow equilization of pressure part of the lab.

Taking averages of the final pressures and temperatures at the marked time ranges on the graphs, similar to Figures 1 and 2, we

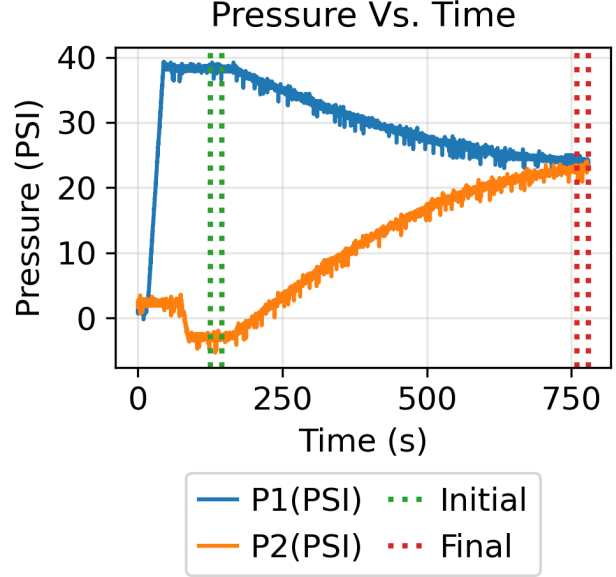


Figure 3: Tank pressures during the Part 1b slow expansion.

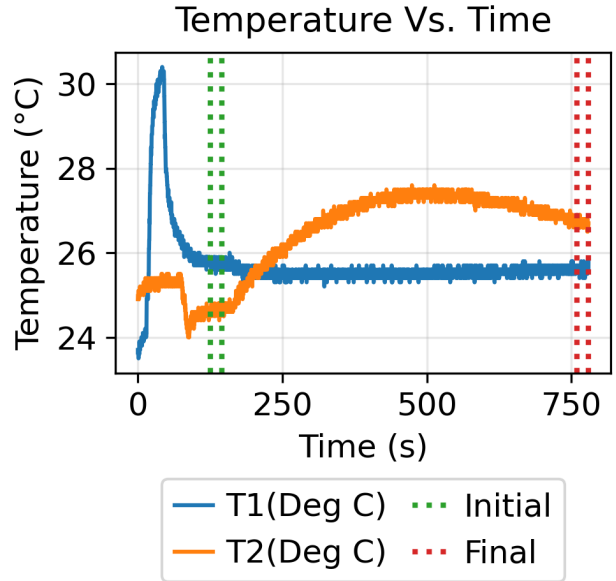


Figure 4: Tank temperatures during the Part 1b slow expansion.

obtain the following values:

$$P_{2R} = 22.94 \text{ PSI}, P_{2L} = 24.04 \text{ PSI}$$

$$T_{2R} = 26.70^\circ\text{C}, T_{2L} = 25.60^\circ\text{C}$$

Then applying the Ideal Gas Law in the same way, we obtain the following:

$$\begin{aligned}\frac{V_L}{V_R} &= \frac{P_{2R}}{P_{2L}} \cdot \frac{T_{2L}}{T_{2R}} \\ \frac{V_L}{V_R} &= \frac{22.94}{24.04} \cdot \frac{25.60}{26.70} \\ \frac{V_L}{V_R} &= 0.915\end{aligned}$$

2.1.2 Heat Transfer and Net Heat Transfer

Though heat transfer is a path-dependent quantity, the net heat transfer in both expansion cases is roughly the same because the initial and final thermodynamic states are approximately identical. However, the heat transfer histories differ significantly between the two processes.

During Part 1A, rapid expansion, once the solenoid valve is opened, the temperature in the left tank experiences a sharp decrease while the temperature of the right tank suddenly increases, before both temperatures slowly reach thermal equilibrium. This sharp change in temperature occurs because the flow of air between the tanks is faster than the rate at which heat can be exchanged with the surroundings.

Meanwhile, in the slow expansion ex-

periment, once the needle valve is opened, the temperatures of the two tanks gradually reach thermal equilibrium over a much longer period of time, without sudden changes in temperature.

When determining the net heat transfer, both tanks can be considered as a single closed system. Because the air initially and finally reaches approximately ambient temperature, there is no boundary work, and the total mass of the system remains constant, the First Law of Thermodynamics gives:

$$\Delta U = Q - W \quad (4)$$

Since the tanks are rigid, no boundary work is performed, and assuming negligible other work interactions:

$$W = 0 \quad (5)$$

Additionally, since the initial and final temperatures are roughly equal, the change in internal energy is approximately zero:

$$\Delta U \approx 0 \quad (6)$$

Therefore, the net heat transfer is:

$$\boxed{Q_{\text{net}} \approx 0} \quad (7)$$

2.1.3 Path Independence of State Properties

As mentioned previously, a thermodynamic state property is a property whose value depends only on the current state of the system and not on the process used to reach that state. The experiment demonstrates the path independence of state properties by comparing two different methods of pressure equalization between interconnected tanks.

In Part 1A, air is transferred rapidly through a solenoid valve, leading to sudden changes in pressure and temperature, as seen in Figures 1–2. Alternatively, in Part 1B, air is transferred gradually through a needle valve, producing slower changes in these properties, as seen in Figures 3–4.

Despite these differences in the paths of the two processes, the final equilibrium pressure is approximately the same when starting from equivalent initial conditions. Therefore, the experiment illustrates how different processes can connect the same initial and final equilibrium states, while thermodynamic state properties remain independent of the path taken between them.

2.2 Part 2: Initial Mass and Volume of the Left Tank

Part 2 of the lab discusses on measuring mass and volume of the system utilizing the mass flow rate and examining the limits of the ideal gas law by measuring the compressibility of gas as pressure increases.

2.2.1 Processed Pressure, Temperature, and Mass Flow Results

Figures 5–6 show the measured tank pressures and temperatures during Part 2.

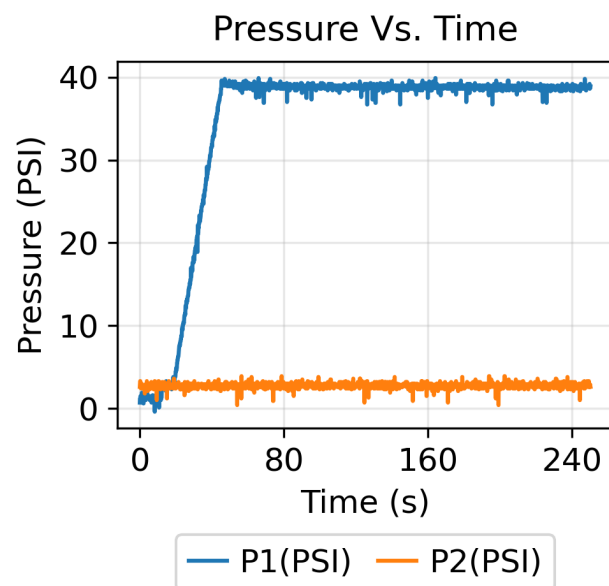


Figure 5: Tank pressures during Part 2.

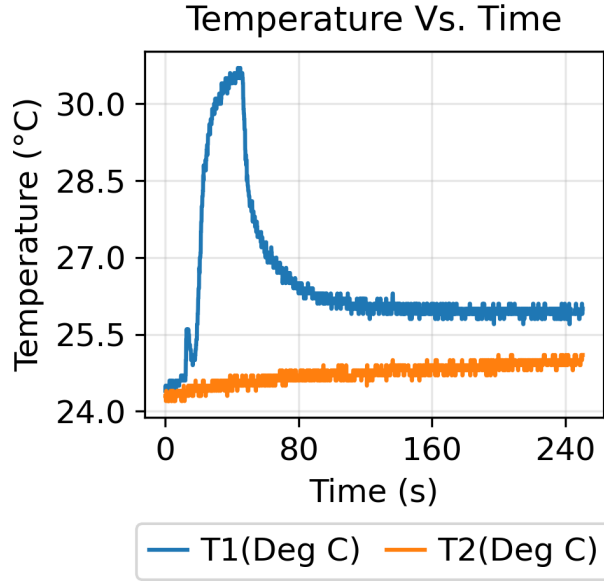


Figure 6: Tank temperatures during Part 2.

2.2.2 Initial Mass and Tank Volume

Figure 7 shows the measured mass flow rate used in the added-mass calculation.

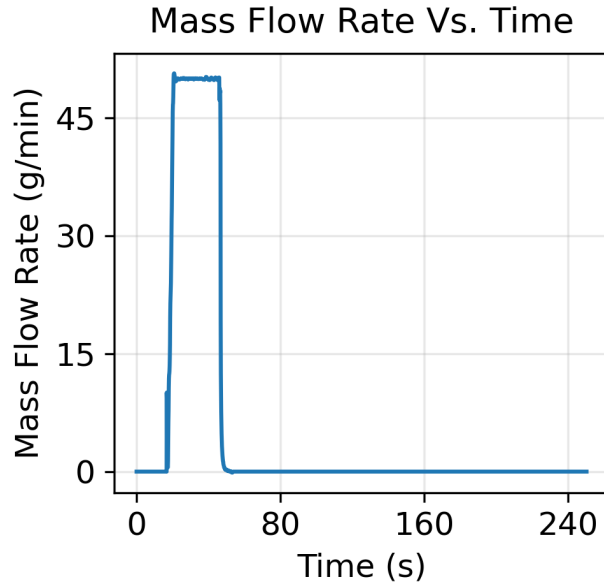


Figure 7: Mass flow rate during Part 2.

Knowing that all added mass is mea-

sured by the sensor, we can calculate the added mass in the left tank as the integral of the mass flow rate - time graph in Figure 7. Using the trapezoid rule with Riemann Sums, we can find that the total mass inside the tank is 23.259g.

Since the mass inside the system after any time is known, we can create a system of equations to calculate V and initial mass.

$$PV = (m_i + \delta m)RT$$

$$P_{\text{gauge}}(30) = 17.6, \quad \delta m(30) = 09.201$$

$$T(30) = 30.1, \quad R = 287$$

$$P_{\text{gauge}}(50) = 38.9, \quad \delta m(50) = 23.259$$

$$T(50) = 28.2, \quad R = 287$$

By using these points and the equation, adjusting for units, I can solve the system of equations and resolve that $V = 0.008201 \text{ m}^3$ or 8.201 L and that the initial mass is 11.712 g

2.2.3 Sources of Error and Limitations

There are several key sources of errors and limitations on this section of the lab, either relating to random human error or systematic error with the apparatus.

When setting the pressure of the system to 40 PSI, we were reliant on a teammate manually checking the gauge pressure. This is limited by human reaction time and uncertainty in the gauge pressure reading, both due to parallax error and uncertainty in measurements. Thus, there was a lot of error with setting the initial pressure to exactly 40 PSI. This is seen through the sensor data which showed that the pressure was slightly below our target of 40 PSI.

Additionally, sensor data displayed that the mass flow rate was not completely constant at 50 g/s, even when temperature and pressure have stabilized for the air in the tank.

2.2.4 Effect of Air Compressibility

The compressibility of air can greatly affect experimental results of pressure and volume, especially as experimental conditions diverge from assumptions made by the ideal gas equation. The ideal gas equation assumes that temperatures are kept high and pressures are kept low. When pressures are kept low, intermolecular forces can be assumed to be negligible. With this model, the com-

pressibility of air can be modelled to be constant. As the pressure increases, electromagnetic repulsions and attractions between particles greatly contribute to the behavior of the gas. This causes the linear model of air compressibility to break down, displaying the inaccuracies in the ideal gas model. Since the intermolecular forces increase as pressure increases, our calculated compressibility is much higher than the actual compressibility.

One can model this using the compressibility factor, where

$$Z = \frac{Pv}{RT}$$

$$\text{or } PV = ZRT.$$

Under constant volume, this measures the divergence of a physical model from Gay-Lussac's Law.

During our experimentation, the maximum gauge pressure found was around 40 psi. Given that this is a similar gauge pressure to that seen in pressurized car tire, this is high enough where our apparatus diverges from assumptions made by the ideal gas model and we should check the compressibility factor during our process.

If we hold the volume V constant, we

can calculate the compressibility factors at low and high pressures.

At $t = 10$:

$$1.3 \cdot v = Z_1 \cdot R \cdot 317.85,$$

$$Z_1 = \frac{1.3 \cdot v}{317.85 \cdot R} = \frac{2}{489} \cdot \frac{v}{R}.$$

At $t = 50$:

$$52.2 \cdot v = Z_2 \cdot R \cdot 301.45,$$

$$Z_1 = \frac{52.2 \cdot v}{301.45 \cdot R} = \frac{1050}{6029} \cdot \frac{v}{R}.$$

Given how the system is held in a rigid tank, R and v can be assumed as constants and thus, it is clear that the ratios between Z_1 and Z_2 is approximately 1:50. Since the compressibility increases by a factor of 50, it is clear that there is nontrivial compression of air within with regime.

Thus, through these calculations, we can see that the compressibility of air significantly increases at higher pressures.

3 Conclusion

Overall, this experiment showed the behavior and limitations of the ideal gas law, along with the behaviours of gasses at different temperatures.

During lab 1, we tested the behavior of mixing gasses with the same initial conditions with both rapid (Part 1a) and slow (Part 1b) expansion. Under this regime, the final conditions of the tank pressures and temperatures in each tank were identical across Part 1a and Part 1b. Through this, we were able to calculate the net heat transfer and prove the path independence of state properties such as heat. Additionally, under both trials, we were able to prove that the volumes of the tanks were near identical.

During lab 2, we were able to calculate the mass of initial air in the tank and volume of the tank given the mass added and measured pressure/temperature values. Additionally, we measured the limitations of the ideal gas model and proved how the ideal gas model breaks down at high pressures, as shown by the compressibility of air. Under the ideal gas model, the compressibility should be constant but under the conditions imposed, the compressibility of air was variable as pressure increased.

References

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- [4] Sanjeev Chandra, *Energy, Entropy and Engines: An Introduction to Thermodynamics*. John Wiley & Sons, 2016.
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